Evaluation for adaptability of Kana-Kanji translation of non-segmented Japanese Kana sentences using inductive learning

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Abstract

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Ordinary Japanese sentences are written using two kinds of characters : i.e. *Kana* and *Kanji*. *Kana* is Japanese phonogramic characters. *Kanji* is ideographic Chinese characters. Therefore we have to use some method in order to input them into the computer. The *Kana-Kanji* translation method is the principal of all input methods. The non-segmented *Kana* input form is especially desirable for users because they do not have to write any segmentations in it. There are many methods for it. These methods still have some problems, which include problems concerning on unregistered words and heavy labor for the making of dictionaries, etc. A new *Kana-Kanji* translation method based on an inductive learning is presented in this paper in order to address those problems. Our proposed method acquires phonogramic and ideographic characters for a word from a *Kanji-Kana* mixed sentence and its non-segmented *Kana* sentence automatically. This method translates a non-segmented *Kana* sentence into the *Kanji-Kana* mixed sentence using the words classified in order of the higher value of the certainty degree. This paper outlines this method and the evaluation experiment for the adaptability of it.

1. Introduction

Ordinary Japanese sentences are written using two kinds of characters : i.e. *Kana* and *Kanji*. *Kana* is Japanese phonogramic characters. *Kanji* is ideographic Chinese characters. There are about fifty *Kana* and several thousand *Kanji*. Independent words are generally written in *Kanji*. Dependent words are written in *Kana*. Table 1 shows examples of a Japanese sentence.

Japanese sentences are not usually written using alphabets. Therefore we have to use some method in order to input them into the computer. The *Kana-Kanji* translation method is the principal of all input methods. The pronunciations of Japanese are able to be expressed by the alphabets as shown in Table 1. Therefore this method translates the pronunciations of Japanese into the

Table 1: Examples of a Japanese Sentence

Kanji-Kana Mixed Sentence		
Japanese	春が来た。	
Pronunciation	*haru ga kita.*	
English	Spring has come.	
Kana		
Japanese	が	た
Pronunciation	*ga*	*ta*
A part of speech	Postposition	Inflectional ending
Kanji		
Japanese	春	来
Pronunciation	*haru*	*ki*
English	Spring	Come
Non-Segmented *Kana* Sentence		
Japanese	はるがきた。	
Pronunciation	*haru ga kita.*	
English	Spring has come.	

Kanji-Kana mixed sentences automatically. There are several kinds of *Kana-Kanji* translation methods. The non-segmented *Kana* input form is especially desirable for users because they do not have to write any segmentations in it. There are many methods for the non-segmented *Kana-Kanji* translation method[Abe 86, Makino 80, Yoshimura 83, Yamashina 88, Araki 87]. These methods are based on heuristics. These methods still have some problems, which include problems concerning on unregistered words and heavy labor for the making of dictionaries, etc. Therefore a new *Kana-Kanji* translation method based on inductive learning is presented in this paper. Our proposed method acquires both pronounced characters and written characters for a word from a *Kanji-Kana* mixed sentence and its non-segmented *Kana* sentence. In this method, words are classified into some ranks based on the certainty degrees. The certainty degree of a word is based on the situation of the acquisition of the word and the rate of the correct translation. And then this method translates a non-segmented *Kana* input sentence into the *Kanji-Kana* mixed sentence using the words classified in order of the higher value of the certainty degree[Araki 91, Takahashi 93]. We have already proposed a morphological analysis method for *Kanji-Kana* mixed sentences by inductive learning and confirmed that it was effective through experiments[Araki 92].

This paper outlines this method and the experimental results for the evaluation of its adaptability. Our proposed method can create dictionaries for *Kana-Kanji* translation automatically. Therefore the system based on this method can adapt to users or the treated fields. We consider that this method is as effective as the text input method by speech because this method can create dictionaries from pronunciations and its *Kanji-Kana* mixed sentences automatically.

2. Outline of This Method

Figure 1 shows an outline of this method. A non-segmented *Kana* sentence is translated into *Kanji-Kana* mixed sentence in order of the higher value of the certainty degree. The certainty degree is based on the situation of the acquisition of the word and the rate of the correct translation. And then human beings correct the translation errors. In the learning process, the system gets word candidates from a non-segmented *Kana* sentence and its translated *Kanji-Kana* mixed sentence using the common parts between them. And then the system gets words by extracting both common and different parts from the word candidates. In the feedback learning process, the system increases the promising degrees of correctly translated words and decreases the promising degrees of erroneously translated words. The system does this by comparing both translated results and its proofread results. Therefore the rate of the correct translation increases gradually by the learning and the feedback learning process even if the initial dictionary is empty.

3. Kana-Kanji Translation Using Inductive Learning Method

3.1 Translation Process

In this method, words are classified into five ranks. The word ranks are MS (the Most certain Segment), CS (Common Segment), S1 (Segment One), RS (Remained Segment) and LS (Less certain Segment). The order of higher certainty degree is MS, CS, S1, RS and LS. Therefore a non-segmented *Kana* sentence is translated in this order. When

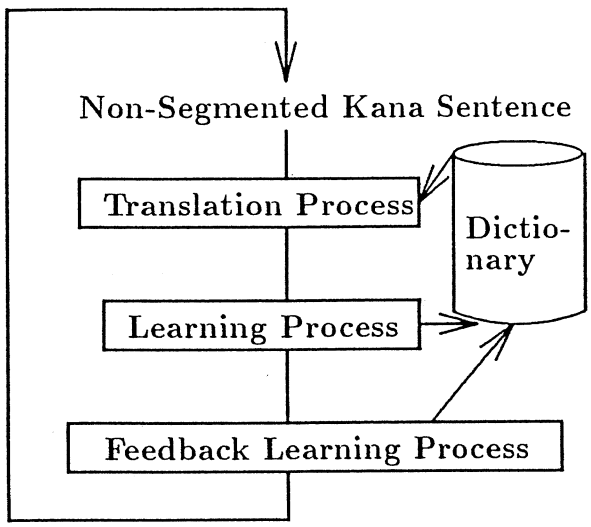


Figure 1: Outline of Process

two characters of the non-segmented *Kana* sentence and the pronounced characters of a word are at least the same, the characters of the non-segmented *Kana* sentence are decided as the word. And then a single-syllable word is decided as the word in the same order of word ranks when both sides of the part are decided as known words.

When there are duplicate word candidates in the same rank, the word candidate which has the most promising degree is decided as the correct word. The promising degree is calculated by the promising evaluation function defined as:

$$PEF = AF + \alpha \times CF - \beta \times EF \qquad (1)$$

when α and β are coefficients, the PEF is the promising evaluation function, AF is the appearance frequency, CF is the frequency of the correct translation and EF is the frequency of the erroneous translation.

3.2 Learning Process

3.2.1 Extraction of Word Candidates

Word candidates are extracted from non-segmented *Kana* input sentences and their translated *Kanji-Kana* mixed sentences using their common parts. This word candidates are called segment one and expressed as S1. The procedures for extraction of S1 are as follows:

1. The common parts between a non-segmented *Kana* sentence and its *Kanji-Kana* mixed sentence are decided.
2. The correspondence of the common parts are decided by the order of the common parts in which they appear in the sentences.
   (a) The different parts between two common parts are extracted.
   (b) The common parts and the different parts are extracted as S1

Table 2 shows examples of the extraction of S1. In Table 2, the underlined parts show common parts between a non-segmented *Kana* sentence and the *Kanji-Kana* mixed sentence.

Table 2: Examples of the Extraction of S1

Non-Segmented *Kana* Sentence	
Japanese	われわれはしゅじゅのほうほうをおこなった
Pronunciation	wareware wa syuzyu no houhou wo okona tta.
English	We have tried a variety of methods.

Translation Result (Kanji-Kana mixed sentence)	
Japanese	我々は種々の方法を行なった。
Pronunciation	wareware wa syuzyu no houhou wo okona tta.
English	We have tried a variety of methods.

Results of the Extraction of S1's		
Japanese	Pronunciation	English
(われわれ:我々)	wareware	We
(は:は)	wa	Postposition
(しゅじゅ:種々)	syuzyu	Variety
(の:の)	no	Postposition
(ほうほう:方法)	houhou	Method
(を:を)	wo	Postposition
(おこな:行な)	okona	Try
(った:った)	tta	Inflectional ending

In "(X:Y)", X is the pronounced character and Y is the written character.

This example of Table 2 has no problems. However the extraction results have errors when the pronunciation of the *Kanji* includes that of postposition. Table 3 shows examples of errors of the

S1 extraction. In this table, the underlined parts show the common parts between a non-segmented *Kana* sentence and its *Kanji-Kana* mixed sentence.

Table 3: Examples of Errors of the S1 Extraction

Non-Segmented *Kana* Sentence	
Japanese	そのすんぽうはびさいかされつつある。
Pronunciation	sono sunpou wa bi retutuarusa ikasa.
English	Its size is decreasing minutely
Translation Result (*Kanji-Kana* mixed sentence)	
Japanese	その寸法は微細化されつつある。
Pronunciation	sono sunpou wa bisaika sa retutuaru.
English	Its size is decreasing minutely

Results of the Extraction of S1's		
Japanese	Pronunciation	English
(その:その)	sono	Its
(すんぽう:寸法)	sunpou	Size
(は:は)	wa	Postpositional
(び:微細化)	(bi : bisaika)	ERROR
(さ:さ)	sa	ERROR
(いかさ:)	(ikasa :)	ERROR
(れつつある :れつつある)	retutuaru	Inflectional ending

In Table 3, "ERROR" represents an error of S1. These errors are made by the left-to-right directional analysis. Table 4 shows the result of the right-to-left directional analysis. All S1's of Table 4 are correct. However the system can not decide whether those are correct or not. Therefore the system decides that the same S1's between those extracted by the left-to-right directional analysis in Table 3 and those extracted by the right-to-left directional analysis in Table 4 are correct. Therefore the S1's registered in the dictionary are three S1's in Table 4. The other two S1's are not registered in it. However these S1's are possible to be extracted from other sentences.

3.2.2 Extraction of CS and RS

The system extracts the common part and the different part from two S1's because an S1 may include another S1. These common parts are called common segment and are expressed by CS. These different parts are called remained segment and expressed by RS. The S1 is deleted from the dictionary when it is divided into CS and RS. Table 5 shows an example of the extraction of CS and RS.

Table 4: Result of Right-to-Left Directional Analysis

Non-Segmented *Kana* Sentence	
Japanese	そのすんぽうはびさいかされつつある。
Pronunciation	sono sunpou wa bisaika saretutuaru.
English	Its size is decreasing minutely.
Translation Result (Kanji-Kana mixed sentence)	
Japanese	その寸法は微細化されつつある。
Pronunciation	sono sunpou wa bisaika saretutuaru.
English	Its size is decreasing minutely.

Results of Extraction of S1's		
Japanese	Pronunciation	English
(その:その)	sono	Its
(すんぽう:寸法)	sunpou	Size
(は:は)	wa	Postpositional
(びさいか:微細化)	bisaika	Minutely
(されつつある: されつつある)	saretutuaru	Inflectional ending

S1's registered in dictionary		
Japanese	Pronunciation	English
(その:その)	sono	Its
(すんぽう:寸法)	sunpou	Size
(は:は)	wa	Postpositional

Table 5: Example of Extraction of CS and RS

S1's	
Japanese	(れんぞくおんせい : 連続音声)
Pronunciation	renzokuonsei
English	Continuous speech
Japanese	(おんせい : 音声)
Pronunciation	onsei
English	Speech
CS	
Japanese	(おんせい : 音声)
Pronunciation	onsei
English	Speech
RS	
Japanese	(れんぞく : 連続)
Pronunciation	renzoku
English	Continuation

Table 6 shows the conditions for the extraction of CS and RS. Condition 1 of Table 6 is based on the difference of information quantity in comparison to a phonogram and an ideograph. Condition 2 of Table 6 is based on the fact that S1 may include some words. The case of $D = 0$ of condition 1 corresponds to phonogram. The case of $D \neq 0$ of condition 1 corresponds to ideograph.

Table 6: Conditions of Extraction of CS and RS

Condition 1 : Conditions for the Number of Characters	
$D = PN - IN$	
Case Condition	$D = 0$ Three written characters are the same at least.
Case Condition	$D \neq 0$ Two written characters are the same at least.
Condition 2 : Situation of Duplication	
S1 includes the other S1.	

"PN" represents the number of pronounced characters of a word.
"IN" represents the number of written characters of a word.

3.2.3 Process for Single-Syllable Word

Table 7 shows an example of the combination process for sequential single-syllable words. "書き込む (fill out)" is one word. However the system divided it into four words "書", "き", "込" and "む" because a single-syllable word is decided as the word only when both sides of it are decided as known words. Therefore this method is added to the process of combining sequential single-syllable words as a word.

3.3 Feedback Learning Process

3.3.1 Update of AF, CF and EF

In this process, the system increases the promising degree of correctly translated words and decreases the promising degree of erroneously translated words. The promising degree is calculated by PEF of the equation (1). In this process, AF, CF and EF are updated. The system adds one to the value of AF for a word when the word is used in the translation process, and also when the word is extracted as S1 in the learning process. The value of AF for CS is the sum of AF's of two S1's from which it was extracted when the S1 is divided into CS and RS. AF for RS is AF for S1 from which it was extracted. The S1 includes RS.

Table 7: Example of the Combination Process for Sequential Single-Syllable Words

Sequential Single-Syllable Words	
Japanese	Pronunciation
(か : 書) (き : き) (こ : 込) (む : む)	ka ki ko mu
Word	
(かきこむ : 書き込む)	kakikomu

The correct translation is defined as one in which the translated result and the proofread result are the same. The erroneous translation is defined as one in which the translated result and the proofread result are different. The system adds one to the value of CF for a word when the word is used for the correct translation. The system adds one to the value of EF for a word when the word is used for the erroneous translation.

3.3.2 Update of Word Ranks

The word ranks are updated using the rate of the correct translation. The correct translation rate is defined as

$$CR = \frac{CF}{CF + EF} \tag{2}$$

when CR is the rate of the correct translation, CF is the frequency of the correct translation and EF is the frequency of the erroneous translation.

Table 8 shows the rules for the update of word ranks. These rules are performed when the sum of CF and EF is more than three[Takahashi 92]. When the CR of a word in LS again reaches more than 40%, the word returns to the original rank.

Table 8: Rules for Update of Word Ranks

Rank	Conditions	
	Extracted Rank	CR
MS	CS	CR $\geq$ 95%
CS	S1	40% $\leq$ CR $<$ 95%
S1	Text	CR $\geq$ 40%
RS	S1	CR $\geq$ 40%
LS	MS, CS, S1 and RS	CR $<$ 40%

4. Experiments on Adaptability

4.1 Data and Procedure

The system based on our proposed method has been developed for evaluating the adaptability of this method. The data used in this experiment includes 10 papers on information processing, 10 papers on mechanical engineering, 10 papers on applied chemistry and 10 papers on cultural science. The total number of data is 40 papers, which consists of 259,187 characters. In this experiment, the values of the coefficients in equation (1) are $\alpha = 2$ and $\beta = 28$. This values are decided by another experiment[Takahashi 92]. The dictionary is empty at first. The procedure of this experiment is Figure 1.

4.2 Results and Considerations

Figure 2 shows the changes in the translation rates. In Figure 2, the rate of the correct translation for the first paper is about 80%. The rate for the last paper on information processing is about 93% using words acquired by inductive learning. However the rate of the correct translation for the first paper on mechanical engineering is under 80%. The reason is that the number of the unregistered words in the dictionary increases because the field of input data changes. However the rate of the correct translation for the last paper on mechanical engineering is again about 90% by inductive learning. This situation is the same on chemical engineering. However the rate of the correct translation for the first paper on cultural science is almost the same as that for the last paper on chemical engineering. It shows that the system adapted to the new field immediately. The reason is that the number of registered words in the dictionary and the number of the characters of the first paper on cultural science are large enough.

5. Conclusions

In this paper, we proposed the new method using inductive learning for *Kana-Kanji* translation of a non-segmented Japanese *Kana* sentence. This method acquires pronounced characters and written characters for a word from a *Kanji-Kana* mixed sentence and its non-segmented *Kana* sentence. Therefore the system based on this method can adapt to users or the treated fields. We developed the experimental system on the basis of this method and have done the experiment to evaluate the adaptability of this method. According to our results, the rate of the correct translation for the adaptability of this method was over 90% for all four fields. These results show that this method is effective for many kinds of fields in the *Kana-Kanji* translation of non-segmented Japanese *Kana* sentences. Future problems for this method are the introduction of heuristics and the decrease of CPU time.

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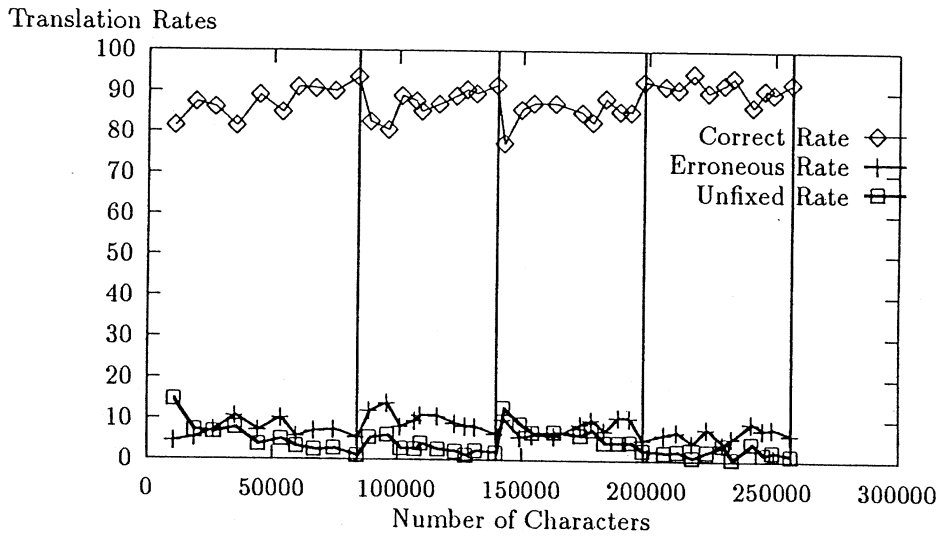


Figure 2: Changes in the Translation Rates

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